MIL-C-46577D
7 December 1973
SUPERSEDING
MIL-C-0046577C
17 April 1972

MILITARY SPECIFICATION

CARTRIDGE, 20MM, ARMOR PIERCING INCENDIARY, M53

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electric primed cartridge for use in Guns, 20mm, Automatic. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

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2.1 (Cont)

DRAWINGS

U . S . ARMY MUNITIONS COMMAND

D7258802	- Cartridge, 20mm, Armor Piercing Incendiary, M53
D7258928	- Marking on Shipping and Storage Container, M548 for 20mm Cartridges
F7258941	- Method of Packing 100 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258942	- Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258968	- Method of Packing 100 Cartridges, 20mm, in M12 Type Links, in M548 Container
F7258969	- Method of Packing 100 Cartridges, 20mm, in M14 Type Links, in M548 Container
F7258970	- Method of Packing 100 Cartridges, 20mm, in M17 Type Links, in M548 Container
D7259237	- Marking on Shipping and Storage Container, M548 for Cartridges, 20mm, Linked (Funtional)
IEL 7567234	- Index of Inspection Equipment Lists for: Cartridge, 20mm, Armor Piercing Incendiary, M53

PUBLICATIONS

U. S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIRED

3.1 General. -The cartridge shall comply with Drawing (Dwg.) D7258802, referenced specifications, and other requirements specified herein.

3.2 First Article Sample. - Unless otherwise directed by the contracting officer, a first article sample shall be required.

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3.3 Manufacturing Process. - All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.3)

3.4 Projectile Extraction.-The force necessary to extract the projectile from the cartridge case shall be as specified on Dwg. D7258802.

3.5 Electrical Resistance. Primed Case. -The electrical resistance of the primed case shall be not less than 500 ohms or more than 1,200,000 ohms.

3.6 Projectile Torque. -The projectile shall withstand a minimum torque as specified by Dwg. D7258802.

3.7 Velocity. -The average velocity of the sample cartridges conditioned at 68 degrees Fahrenheit (°F) to 72°F shall be 3380 ± 50 feet per second (fps) at 78 feet from the test weapon muzzle and the sample standard deviation shall not exceed 40 fps.

3.8 Pressure.-The average chamber pressure (X) of the sample cartridges, conditioned at 68°F to 72°F, plus 0.84 times the average range (R) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

3.9 Waterproofness. -The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F to 72°F shall not vary from the average velocity (dry) by more than 100 fps.

3.10 Accuracy. -The average mean radius of all targets of the sample cartridge projectiles at ambient temperature shall not exceed 15 Inches at a range of 600 yards.

3.11 Action Time. -The action time of the cartridge shall not exceed 4.0 milliseconds at either ambient or - 65°F temperature.

3.12 Incendiary Performance.-The projectile shall function with acceptable incendiary characteristics upon target impact.

3.13 Drop Sensitivity. -The projectile shall not function on impact upon a steel plate at a velocity of 52 ± 5 fps.

3.14 Function and Casualty. -The cartridge, at ambient temperature, shall function without casualty.

3.15 Workmanship .-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

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3.15.1 Process Defects. -The cartridge shall be free of defects such as folds, wrinkles, buckles, cracks, splits, severe scratches, cuts, dents, perforations, loose components and other forms of substandard workmanship.

3.15.2 Cleanliness. -The cartridge and its components shall be free of all foreign matter. (see 6.6)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such Inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3).
2. Quality Conformance Inspection (see 4.4).

4.3 First Article Inspection.

4.3.1 First Article Sample.-At the beginning of production, a sample representative of the production material, equipment, processes and procedures shall be submitted in accordance with contract requirements and shall consist of 695 cartridges. Prior to approval of the first article sample, the acquisition of material or components for, or the commencement of production of the balance of the contract quantity shall be at the sole risk of the contractor.

4.3.1.1 First Article Inspection.-Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings and specification requirements. The contractor's inspection shall be subjected to verification by the Government representative.

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4.3.1.2 Examination and Tests. - The first article sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such facility specified in the contract. Determination as to approval of any first article sample shall be based only only upon results of initial examinations and tests.

4.3.1.3 First Article Sample Failure. -Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.4 Quality Conformance Inspection.

4.4.1 Submission of Product. -The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.4.1.2 Lot Identification. -Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a federal stock number assigned by the procuring activity.

4.4.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

4.4.2.1 Primed Cartridge Cases. -Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.2.1. Primed cases which fail to meet the requirements of 3.5 shall be rejected.

4.4.2.2 Sampling and Acceptance Criteria. -The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually.

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4.4.2.2 (Cont)

However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.1.c) In addition where MIL-STD is used, sampling plans applicable to a class shall not be used for major defects. (see 6.5)

4.4.2.2.1 AQL's. -The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.257%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

4.4.3 Classification of Defects 1/ . - The classification of defects shall be aS follows:

4.4.3.1 Charged nose assembly (see Dwg. C7258806 and Dwg. D7258803, a detail of Dwg. D7258802.

Category and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Length, shoulder to top of charge	Gage
Minor: None defined.	

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4.4.3.2 Cartridge Assembly (see Dwg. D7258802 and Dwg. D7258803, a detail of Dwg. 1)7258802).

	Category and Defects	Method of Inspection	
Critical			
1.	Split or perforated case	Visual	
2.	Crack or split in projectile	Visual	
3.	Crimp missing (case-projectile)	Visual	
Major:			
101.	Loose primer	Manual	
102.	Depth of primer seating	Gage	
103.	Over-all length, max.	Gage	
104.	Improper or incomplete crimp (case-projectile)	Visual	
105.	Profile and alignment, max.	Gage	2/
106.	Corrosion	Visual	
107.	Head configuration	Visual	
108.	Metal sliver on head face	Visual	3/
109.	Length, case head to centerline of crimp	Gage	
110.	Inverted primer	Visual	
Minor:			
201.	Marking incorrect, incomplete, illegible or missing	visual	
202.	Projectile protective coating damaged or incomplete	Visual	
203.	Waterproofing missing around primer	Visual	
204.	Scratch, dent, buckle, bulge, wrinkle or fold in case	Visual	
205.	Rotating band damaged	Visual	
206.	Primer crimp missing	Visual	
207.	Foreign matter, except corrosion	Visual	
208.	Incorrect type cartridge or component	Visual	
209.	Lacquer on primer button	Visual	
210.	Gap between nose and adaptor	Gage	

1/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

3/ A metal sliver bridging the primer insulator shall be cause for classifying the cartridge defective.

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4.4.4 Tests. -The tests listed in Table I shall be perfomed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Sample Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the test sample is representative of the entire lot shall be used in the tests. The selected test sample cartridges shall be thoroughly mixed before being divided into samples for the various tests. The combining of tests is permitted. (see 6.9)

Table I

Test 1/		Sample Size Production Lot	Rqmt Para
Projectile extraction	2/	20	3.4
Electrical resistance verification	3/	20	3.5
Propellant contamination	4/	20	3.15.2
Projectile torque	5/	20	3.6
Velocity	6/	20	3.7
Pressure	7/	20	3.8
Waterproofness	8/	20	3.9
Accuracy	9/	50	3.10
Action time	10/		3.11
Ambient temperature		25	
-65°F temperature		25	
Incendiary performance	11/	20	3.12
Drop sensitivity	12/	32	3.13
Function and casualty	13/	400	3.14

1/ The lot shall be rejected if in any firing test, one or more of the following malfunctions or casualties of Table II occur:

a. Misfire associated with e or f of Table II of Note 2/.
b. Projectile remaining in bore.
c. Metal parts separation, except rotating band separation (see Table II, Note 5/).
d. Premature functioning of the projectile prior to target impact.

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

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2/ Projectile Extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to comply with the maximum requirement.

3/ Electrical Resistance Verification. -The lot will be rejected if any primer of the sample cartridges fails to comply with the primer resistance requirement.

4/ Propellant Contamination. -The presence of foreign matter in the propellant of one or more of the test cartridges will cause rejection of the lot. (see 3.15.2 and 6.6)

5/ Projectile Torque. -Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot . If one cartridge of the sample fails to comply a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

6/ velocity. -If the sample fails to comply with either ox both requirements, a retest sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to comply with the applicable requirements. (see 6.7.1)

7/ Pressure. -If-the average pressure $(\bar{X})$ plus 0.84 times the average range (R) of the sample cartridges exceeds 60,500 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure $(\bar{X})$ plus 0.91 times the average range $(\bar{R})$ of the retest sample fails to comply with the applicable requirement.. (see 6.7.2)

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8/ Waterproofness. -If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.7.

9/ Accuracy. -If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 accuracy targets exceeds 15 inches at the same range.

10/ Action Time. -If any cartridge of the sample fails to meet the requirement or if the sample average action time (X) plus 4 sample standard deviations (S) exceeds 4 milliseconds, the lot shall be rejected. These criteria shall apply separately to both the ambient and cold phase tests. (see 6.7.1)

11/ Incendiary Performance. -Failure of three or more sample cartridges to comply with the requirement shall cause rejection of the lot. If one or two cartridges fail to comply with the requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if four or more cartridges of the combined samples fail to comply with the requirement.

12/ Drop Sensitivity. -Failure of one or more sample cartridges to comply with the requirement shall cause rejection of the lot.

13/ Function and Casualty. -The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number (s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 800 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

4.4.4.1 Firing defects. -Firing defects and associated criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

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4.4.4.1 (Cont)

Table II

Firing Casualties	Acc	Rej
Cartridge:		
Fire "out of battery," M61 gun 1/	0	1
Misfire 2/	-	
Failure to chamber	0	1
Failure to extract	0	1
projectile remaining in bore	0	1
Primer:		
Missing primer or button missing from primer	0	2
Primer leak	15	40
Case:		
Longitudinal split 3/		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture, partial 3/		
S, J or K	4	11
L	0	1
Circumferential rupture, complete 3/	0	1
Detached metal 4/	0	1
Projectile:		
Rotating band separation 5/		
Complete or partial separation having band seat serration imprint	0	3
Separation having length greater than 0.5 inch, and without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 6/	0	1

1/ A cartridge shall be classed defective if it fires, due to delayed ignition after the bolt has been unlocked and extraction has been initiated or completed.

2/ Misfiring test cartridges, excepting those from the waterproofness test (Table I), shall be retested in accordance with the method of 4.5.11. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

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4.4.4.1 (Cont)

2/ (Cont)

a. Excessive primer seating depth.
b. Metal slivers across primer insulator.
c. Electrical continuity.
d. Missing or obstructed case vent.
e. Missing or insufficient primer composition or propellant.
f. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "Acc" number shall be zero (0) and the "Rej" number 3, for misfired associated with conditions b, c, or d. The second sample for evaluation of misfires associated with conditions b, c, or d shall consist of the function and casualty second sample (see Table I, Note 13/). The lot shall be rejected if either condition e or f is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

3/ See Dwg. C7258801, a detail of IEL 7567234, for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table 11 for the areas involved shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

4/ Metal sheared or missing from the fired cartridge case exterior, such as, rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

5/ For classification as a defect, there must be evidence either by recovery of the band or portion thereof, or by hole(s) in the fragmentation screens. The lot shall not be penalized for normal band fringing or the separation of band fragments having no evidence of band seat serration imprint and length not exceeding 0.5 inch.

6/ Breakup or separation of projectile metal part(s), except rotating band, as witnessed by recovery of the part(s) or fragment(s), or by hole(s) in the fragmentation screens, shall be classed as a defect.

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4.4.4.2 Unlisted Firing Defects. -The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test, indicates that the product is unsuited for the purpose intended.

4.4.5 Packaging, Packing and Marking Inspection. - Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.4.6 Inspection Equipment. -Index of Inspection Equipment List No. IEL 7567234 identifies the applicable Inspection Equipment List required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, see Dwg. B11075228, a detail of IEL 7567234).

4.5 Test Methods and Procedures.

4.5.1 Projectile Extraction.-The method of test shall be as prescsibed by AMCR 715-505, Volume 8.

4.5.2 Electrical Resistance.

4.5.2.1 Primed Cartridge Cases. -The equipment and circuitry specified by IEL 7567234 shall be used to measure the resistance of the primer In each primed cartridge case.

4.5 .2.2 Electrical Resistance Verification. - A Simpson Volt-Ohm-Ammeter, Model 260, or approved equal, shall be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX 100 and RX 100 and RX 10,000 scales of the meter will be used in order limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum.

4.5.3 Propellant Contamination.-The propellant from each of the cartridges tested for projectile extraction shall be examined visually for contamination, as prescribed by AMCR 715-505, Volume 8.

4.5.4 Projectile Torque.-Equipment shall be as specified by IEL 7567234. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque as specified by 3.6 shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalign-ment in the scratch mark, shall be recorded.

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4.5.5 Velocity. Pressure and Waterproofness. -The methods of test shall be as prescribed by AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed. (see 6.7)

4.5.6 Accuracy. -The method of test and measurement of targets shall be as prescribed by AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.5.7 Incendiary Performance. - The method of test and grading of incendiary performance shall be as prescribed in AMCR 715-505, Volume 8. The projectile performance shall begin within one foot after impact against .040 inch nominal thickness aluminum plate (2024T3, T4 or T361) normal to line of fire located at 200 ± 25 yards from muzzle of the gun.

4.5.8 Drop Sensitivity. -The method of test shall be as prescribed in AMCR 715-505, Volume 8. The nose of the projectile shall impact the steel plate, normal to projectile axis, at a velocity of 52 ± 5 fps.

4.5.9 Action Time.-The method of test shall be as prescribed by AMCR 715-505, Volume 8. Statistics necessary for testing of results against the acceptance criterion shall be computed. (see 6.7.1)

4.5.10 Function and Casualty. -The method of test shall be as prescribed by AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges and shall be divided and fired in equal numbers in the M39 and M61 weapons. The weapon barrel(s) shall be at ambient temperature at beginning of test and cooled to ambient after each 100 rounds. rate of each burst shall not be less than 1400 shots per minute (SPM) in the M39 weapon and 5500 SPMin the M61 weapon. burst shall be recorded.

4.5.11 Misfires.-Misfiring cartridges shall be retested in the test equipment prescribed by the IEL for the accuracy test. Results of examinations required by Table II shall be recorded.

4.5.12 Defective Weapon. -If any test weapon fails to comply with the minimum cyclic rate established for the weapon, or any test cartridge firing defect is found to have been caused by a defective test weapon, the test shall be suspended.shall be corrected or replaced and the test repeated in whole or in part as indicated. If the test cartridge defect is not found to have been caused by the defective weapon, it shall be valid for lot penalty in accordance with Table II.

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5. PREPARATION FOR DELIVERY

5.1 Packing, Level A. - The type of ammunition pack shall be in accordance with instructions from the procuring activity. Linked ammunition shall be packed in accordance with Dwg. F7258968, F7258969, or F7258970. Bulk ammunition shall be packed in accordance with Dwg. F7258941 or F7258942.

5.2 Marking. -Marking on the shipping and storage container shall be in accordance with Dwg. D7258928 or D7259237, as applicable.

6. NOTES

6.1 Intended Use. -These cartridges are intended for use in various automatic gun systems that have been designed for firing cartridges with the M103 type cartridge case configuration.

6.2 Ordering Data. -Invitation for bids and contracts or orders till specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 First article sample requirements. (see 3.2, 4.3 and 6.4)

6.2.4 Packing and marking instructions. (see 5.1 and 5.2)

6.2.5 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance Inspection purposes.

6.2.6 Provision for the submission and approval of the manufacturing process.

6.2.7 Instructions for shipment of first article end production lot samples for firing tests.

6.3 Process Deviation. -A process deviation is defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 First Article Sample.-In order to minimize inspection costs, the firing tests of 4.4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the testing facility.

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6.5 AQL's. -The optional use of AQL value. for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.6 Contamination of Explosive Components. -Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfires, increase action time and lot rejection.

6.7 Computations.

6.7.1 Standard Deviation. -Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$S = \sqrt{\frac{(X_i - \bar{X})^2}{n}}$$ or equivalent

6.7.2 Range Plan.-Where a range plan is specified, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range (R).

6.8 International Agreements.-Certain provisions of this specification are the subject of international standardization agreement STANAG 3585. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.9 Combing test. - With approval of the procuring activity, tests may be performed concurrently on the same sample cartridges provided that the test results are not affected by this procedure. The test facility should make maximum use of this-procedure to minimize testing costs.

NOTE: Asterisks not used in this revision to identify changes with respect to the previous issue, due to hte extensiveness of the changes.

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Custodian:

Army - MU
Air Force - 70

Preparing Activity:

Army - MU

Reviewers:

Army - MU
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